

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience turning large volumes of financial data into audit-relevant insights at Deutsche Bundesbank (AnaCredit), where I built parameterized validation frameworks and statistical anomaly detection for regulatory reporting data. Comfortable working with Python, SQL, and BI tooling to support data-driven audit approaches, and translating technical findings for non-technical stakeholders.

Core Competencies

- **Programming:** Python, SQL, R, Java, C++
- **Data-Driven Audit Support:** Parameterized validation frameworks for large-scale financial/reporting data; statistical anomaly and outlier detection
- **Data Quality & Automation:** Automated migration and error-detection testing (Pandas), YAML-driven batch validation, automated stakeholder communication
- **Databases & BI Tools:** SQL, SQLite, MySQL, Tableau, Excel
- **Machine Learning:** Scikit-Learn, Random Forest, SVM, K-Means, predictive modeling
- **Analysis & Communication:** Structured reporting for non-technical stakeholders, German (C1) and English (C1)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticketing and emails to reporting banks
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Automated Validation Framework for Regulatory Reporting Data – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution analyzing large regulatory databases

 - Designed parameterized MySQL queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Extended the validation logic with percentile-based statistical benchmarking over a rolling 12-month window to flag anomalous reporting frequencies and separate false positives from genuine data-quality issues
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call

Test Data Generator for the Bundesbank Platform – configuration-driven system producing complete, consistent synthetic test datasets to verify internal processing systems

 - Generated Cartesian-product attribute combinations per table with consistent IDs across tables to preserve referential integrity
 - Exported datasets as XML with full schema metadata and plausibility metrics

Projects

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log documenting every change (table, column, old/new value, action, timestamp), plus 5 quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data to identify risk profiles and resource bottlenecks; built predictive models (Random Forest, Neural Networks) and K-Means clustering

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, numerical optimization, cryptography & security.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Certifications

- **Tableau Essentials** – LinkedIn Learning (05/2026).
- **Data Science in Python** – University of Michigan, Coursera (04/2026).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.